

CA-C30L: Mobile Application Development Lab

LAB PROGRAM-12 : GOOGLE MAPS

12. Display Map based on the Current/given location.

Program:

Step 1 : Add Google maps activity use corresponding xml file (instead of regular Mainactivity and activity_main xml files)

- i) Start a new project
- ii) Add a new GoogleMaps activity ; (right click **java folder->New->GoogleMaps Views Activity**)

A new googlemaps activity with corresponding xml file is generated with a ready made code

Step 2 : Generate the API Key

For security and authentication purposes to use google maps ,we need to generate an API key for our project .

From the manifest file ,we get a link in the comment of the automatically generated code :

- i) Copy the URL(<http://console.cloud.google.com>) in the browser address bar and create your account
- ii) A project will be created by default
- iii) Click on Navigation menu (top left) and select APIs and Services-> Credentials
- iv) API key is generated ,copy this key in manifest file as explained further

Step 3 : Set the permissions in manifest file and copy the API key value in metadata of manifest file

- i) *After <manifest> start tag ends ,before <application> tag starts*

```
<uses-permission  
android:name="android.permission.ACCESS_FINE_LOCATION" />  
<uses-permission  
android:name="android.permission.ACCESS_COARSE_LOCATION" />  
<uses-permission android:name="android.permission.INTERNET" />  
ii)    inside application tag ,before <activity> starts add  
    <meta-data  
        android:name="com.google.android.geo.API_KEY"  
        android:value="YOUR_API_KEY" />
```

Step 4 : In MapsActivity.java ,change the latitude and longitude values as desired and change the marker name as you want to be displayed (you can get the latitude and longitude values for your location by right clicking on the position in google maps.)

Note : No changes required in activity_map.xml or MainActivity.java. You can delete them after making the **MapsActivity.java as launcher activity in manifest file using intent tag :**

```
<activity  
    android:name=".MapsActivity"  
    android:exported="true">  
    <intent-filter>  
        <action android:name="android.intent.action.MAIN" />  
  
        <category android:name="android.intent.category.LAUNCHER" />  
    </intent-filter>  
</activity>
```

MapsActivity.java

The rest code except the latitude,longitude lines remains same

```
public class MapsActivity extends FragmentActivity implements  
OnMapReadyCallback {
```

```
    private GoogleMap mMap;  
    private ActivityMapsBinding binding;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    binding = ActivityMapsBinding.inflate(getLayoutInflater());
    setContentView(binding.getRoot());
    SupportMapFragment mapFragment = (SupportMapFragment)
getSupportFragmentManager().findFragmentById(R.id.map);
    mapFragment.getMapAsync(this);
}
```

```
@Override
public void onMapReady(GoogleMap googleMap) {
    mMap = googleMap;

    // Add a marker in India and move the camera latitude and
    longitude -12.911210, 77.610083
    LatLng india = new LatLng(12.911210, 77.610083);
    mMap.addMarker(new MarkerOptions().position(india).title("I
am here"));

    mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(india,
15));
}
}
```

Activity_maps.xml

```
<androidx.fragment.app.FragmentContainerView
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:map="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/map"

android:name="com.google.android.gms.maps.SupportMapFragment"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MapsActivity" />
```
